

IT Access Transition Project

Executive Director Role Handover — Microsoft 365 Environment

Case Study: End-to-end access audit, replication, and verification for a leadership role transition

1. Overview

A mid-sized nonprofit organization needed to onboard a new Executive Director (ED) and ensure their Microsoft 365 access matched that of the outgoing ED exactly — spanning security groups, shared mailboxes, SharePoint sites, and Microsoft Teams. This document summarizes the request, the approach taken, the tools used, and the outcome.

Note: all organization names, personal names, domains, and identifiers in this document have been replaced with generic placeholders for confidentiality.

2. Original Request

The request was received from organizational leadership via email:

"As our new Executive Director will be joining us shortly, could you please create a new email account for her? Before finalizing the email address and login, if you have any recommendations for the email ID based on best practices, please let us know."

"In addition, could you please ensure that her account is configured with the same permissions and access that the previous Executive Director had in his role? This includes all relevant Microsoft 365 access, shared mailboxes, security groups, Teams, and any other permissions associated with the Executive Director position."

"Could you also add her to the appropriate SharePoint sites and grant the same SharePoint permissions the previous ED had?"

3. Objective

- Audit every category of Microsoft 365 access held by the outgoing ED
- Replicate that access on the incoming ED's account, category by category
- Verify parity with a direct comparison, rather than assuming success
- Flag and resolve any access that appeared to be personal/leftover rather than tied to the role itself
- Produce a clean audit trail for compliance and future reference

4. Methodology

4.1 — Audit the Outgoing ED's Access

Using PowerShell with the Microsoft Graph, Exchange Online Management, PnP PowerShell, and Microsoft Teams modules, each access category was independently pulled and exported to CSV:

- Security & Microsoft 365 group memberships (Microsoft Graph)
- Entra ID directory / admin roles (Microsoft Graph)
- Assigned licenses (Microsoft Graph)
- Shared mailbox permissions — Full Access, Send As, and Send on Behalf checked independently, since they are separate, non-overlapping grants (Exchange Online)

- Direct SharePoint site access, checked across every site in the tenant (PnP PowerShell)
- Microsoft Teams membership and role — Owner vs. Member (Microsoft Teams module)

4.2 — Apply Access to the Incoming ED

Rather than manually recreating each grant, the audit CSVs were fed directly back into PowerShell as the source of truth:

- Group memberships were bulk-applied via Microsoft Graph, looping through the exported list
- Shared mailbox permissions were replicated exactly (Full Access)
- SharePoint access was applied by dynamically detecting each site's actual "Members" group rather than guessing group names from the site URL — an early approach that only matched 1 of 17 sites until this was corrected
- Teams membership and role (Owner/Member) were applied to match the outgoing ED precisely

4.3 — Verify Parity

Every category was re-audited against the new ED's account after changes were applied, then compared side by side against the original outgoing-ED export using PowerShell's Compare-Object, rather than assuming the "apply" step succeeded. This surfaced two real discrepancies (see Section 6) that would otherwise have gone unnoticed.

5. Technical Environment

Component	Tool / Module
Identity & Groups	Microsoft Graph PowerShell SDK
Mailbox Permissions	Exchange Online Management module
SharePoint	PnP PowerShell (custom Entra app registration for auth)
Teams	MicrosoftTeams PowerShell module
Shell	PowerShell 7 (cross-platform)

6. Findings & Decisions

6.1 — Access Excluded by Design

One SharePoint site (an internal app management site) showed the outgoing ED as a listed user, but had no site-level groups at all — access there was tied to a broader administrative role rather than the ED position itself. This was confirmed by checking group membership directly (returned empty) before deciding to exclude it, rather than replicating it by default.

6.2 — Role Mismatch Caught by Verification

During verification, one Team showed the new ED assigned as Owner when the outgoing ED had only held Member status. This was corrected to match exactly. This discrepancy would not have been caught without the explicit before/after comparison step — reinforcing that verification is a required step, not an optional one.

6.3 — Pre-existing Baseline Access

The comparison also surfaced group and Team memberships the incoming ED already held prior to this process (standard onboarding baseline). These were left untouched, since the goal was matching the ED-specific access, not overwriting her existing account.

7. Results Summary

Category	Result	Notes
Security & M365 Groups	10 / 10 matched	Bulk-applied via Microsoft Graph
Shared Mailboxes	1 / 1 matched	Full Access permission replicated
SharePoint Sites	16 / 17 matched	One site intentionally excluded — see Findings
Microsoft Teams	7 / 7 matched	Owner/Member roles corrected to match exactly
Entra Directory Roles	N/A	Outgoing user held none — confirmed, not assumed
Licenses	Deferred	Assigned separately via admin center by request

8. Recommendations for Future Transitions

Person-to-person access cloning is thorough but time-intensive, and depends on the outgoing employee's account still existing and being accessible. The recommended long-term improvement is:

- Create a single role-based security group (e.g., "Role-ExecutiveDirector") and point every ED-specific mailbox permission, SharePoint grant, and Teams membership at that group instead of the individual
- Future transitions become: add the new employee to the role group, remove the outgoing employee — a two-step change instead of a multi-category audit
- This also improves auditability, since "who has ED-level access" becomes a single query instead of a reconstructed history

9. Appendix: Key Commands Used

The commands below are sanitized — organization domain, tenant name, user identifiers, and app registration IDs have been replaced with generic placeholders (e.g. organization.org, tenant-admin.sharepoint.com, <client-id>).

9.1 — Connect to Services

```
Connect-MgGraph -Scopes "User.Read.All","Group.Read.All","Directory.Read.All"
Connect-ExchangeOnline -Device
Connect-PnPOnline -Url "https://tenant-admin.sharepoint.com" -Interactive -ClientId "<client-id>"
Connect-MicrosoftTeams
```

Optional — only needed once per tenant, to register an app for SharePoint sign-in (Microsoft retired the shared multi-tenant login option in 2024):

```
Register-PnPEntraIDAppForInteractiveLogin -ApplicationName "PnP PowerShell" -Tenant "tenant.onmicrosoft.com" -Interactive
```

9.2 — Audit: Pull Outgoing User's Access

Group memberships:

```
Get-MgUserMemberOf -UserId $oldUser.Id -All | ForEach-Object {
    Get-MgGroup -GroupId $_.Id | Select DisplayName, Mail, GroupTypes, Id
} | Export-Csv "OldUser-Groups.csv" -NoTypeInfo
```

Directory (admin) roles:

```
Get-MgUserMemberOf -UserId $oldUser.Id -All |
    Where-Object { $_.AdditionalProperties["@odata.type"] -eq
"#microsoft.graph.directoryRole" }
```

Assigned licenses:

```
Get-MgUserLicenseDetail -UserId $oldUser.Id | Select SkuPartNumber, SkuId
```

Shared mailbox permissions — checked as three independent grant types:

```
Get-MailboxPermission -Identity $mbx | Where-Object { $_.User -like "*olduser*" }      #
Full Access
Get-RecipientPermission -Identity $mbx | Where-Object { $_.Trustee -like "*olduser*" } #
Send As
(Get-Mailbox -Identity $mbx).GrantSendOnBehalfTo                                     #
Send on Behalf
```

Direct SharePoint site access, checked across every site in the tenant:

```
$sites = Get-PnPTenantSite | Select -ExpandProperty Url
foreach ($site in $sites) {
    Connect-PnPOnline -Url $site -Interactive -ClientId "<client-id>"
    Get-PnPUser | Where-Object { $_.Email -like "*olduser*" }
}
```

Teams membership and role:

```
Get-Team | ForEach-Object {
    Get-TeamUser -GroupId $_.GroupId | Where-Object { $_.User -like "*olduser*" }
}
```

9.3 — Apply: Grant Access to Incoming User

Bulk-add to all groups from the audit export:

```
Import-Csv "OldUser-Groups.csv" | ForEach-Object {  
    New-MgGroupMember -GroupId $_.Id -DirectoryObjectId $newUser.Id  
}
```

Shared mailbox — Full Access:

```
Add-MailboxPermission -Identity $mbx -User $newUser -AccessRights FullAccess -AutoMapping  
$true
```

SharePoint — dynamically detect each site's Members group rather than assuming a fixed naming pattern:

```
$memberGroup = Get-PnPGroup | Where-Object { $_.Title -like "*Members*" } | Select -First 1  
Add-PnPGroupMember -LoginName $newUser -Group $memberGroup.Title
```

Teams — add with matching role:

```
Add-TeamUser -GroupId $team.GroupId -User $newUser -Role $_.Role
```

Optional — changing an existing member's role (no direct "Set" cmdlet exists; remove and re-add):

```
Remove-TeamUser -GroupId $team.GroupId -User $newUser  
Add-TeamUser -GroupId $team.GroupId -User $newUser -Role Owner
```

9.4 — Verify: Compare Old vs. New

```
Compare-Object (Import-Csv "OldUser-Groups.csv") (Import-Csv "NewUser-Groups.csv") -  
Property Id |  
    Where-Object { $_.SideIndicator -eq "<=" }
```

An empty result confirms full parity; any remaining rows indicate access still missing on the new account.

9.5 — Optional / Situational Commands

Temporary elevated access, used only to read SharePoint site membership during the audit (removed afterward):

```
Set-PnPTenantSite -Url $site -Owners $adminUPN
```

Module troubleshooting — forcing a specific module version when multiple versions conflict:

```
Import-Module Microsoft.Graph.Authentication -RequiredVersion 2.38.0
```

Confirming an active session before assuming a connection failed silently:

```
Get-MgContext  
Get-ConnectionInformation  
Get-PnPConnection
```

10. Outcome

The incoming Executive Director's account was verified to match the outgoing ED's access across groups, shared mailboxes, SharePoint, and Teams, with two real discrepancies identified and corrected through direct verification rather than assumption. One SharePoint access grant was deliberately excluded after investigation showed it was unrelated to the ED role. License assignment was completed separately through the Microsoft 365 Admin Center by organizational preference. A temporary elevated permission used during the SharePoint audit phase was scheduled for removal as a follow-up cleanup step.